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Department of Physics and Astronomy “A. Righi”

Theoretical Physics

Quantum Many Body Theory

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Introduction

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1 | Weakly Interacting Bose Gas

1.1 | The Weak Interaction Assumption

The weakly interacting Bose gas serves as a foundational paradigm in quantum many-body theory. We consider a system of N identical bosons of mass m confined to a box of volume $V = L^3$ with periodic boundary conditions. The interparticle interaction is characterized by a two-body potential $U(\mathbf{r})$, which we assume to be weak. The constituents of this gas obey Bose-Einstein statistics and are treated as non-relativistic elementary particles, meaning that their internal degrees of freedom are not activated at the relevant energy scales, and the total particle number is strictly fixed (precluding, for example, a gas of photons).

TODO: Add a figure of the box with periodic boundary conditions

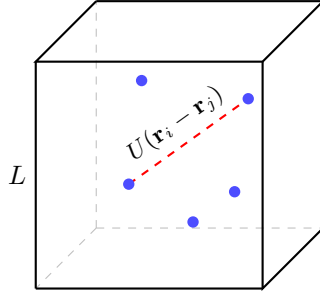


Figure 1.1: Schematic representation of a system of N interacting bosons confined within a cubic box of volume $V = L^3$. The periodic boundary conditions simulate a bulk system. The dashed red line illustrates the two-body interaction potential $U(\mathbf{r}_i - \mathbf{r}_j)$ acting strictly between pairs of particles.

Typical experimental realizations involve dilute gases of bosonic atoms, meaning atoms possessing an even total number of constituent fermions (protons, neutrons, and electrons), such as Sodium (^{23}Na), Rubidium (^{87}Rb), Potassium (^{41}K), Cesium (^{133}Ce), and Lithium (^7Li). Historically, this theoretical framework was first motivated by efforts to explain the superfluid behavior of Helium-4 below 2.17K, despite the fact that liquid ^4He actually resides in a strongly interacting regime.

The Hamiltonian governing this system of N interacting identical bosons is given in first quantization by

$$\hat{H} = \sum_{i=1}^N \frac{\hat{\mathbf{p}}_i^2}{2m} + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j),$$

where $\hat{\mathbf{p}}_i$ is the momentum operator of the i -th particle, and $U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j)$ represents the interaction energy between the i -th and j -th particles.

For a system comprising a macroscopic number of identical bosons, it is highly advantageous to transition to the second quantization formalism.¹ We introduce bosonic creation and annihilation

¹For a very clear introduction to the second quantization formalism for many-particle systems, see: H. Tasaki,

operators associated with the momentum eigenfunctions of a single particle. For a box with periodic boundary conditions, these single-particle states are quantized plane waves:

$$\phi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}},$$

where the allowed wave vectors $\mathbf{k}$ take the discrete values

$$\mathbf{k} = \frac{\pi}{L}(n_x, n_y, n_z), \quad n_x, n_y, n_z \in \{0, \pm 1, \pm 2, \dots\}.$$

In this momentum basis, the many-body Hamiltonian becomes

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_k \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} U(\mathbf{q}) \hat{a}_{\mathbf{k}+\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}'-\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{k}}. \quad (1.1.1)$$

Here, ϵ_k denotes the non-interacting energy associated with the wave vector $\mathbf{k}$ (which depends exclusively on its magnitude k), given by the free-particle dispersion

$$\epsilon_k = \frac{\hbar^2 k^2}{2m},$$

and $U(\mathbf{q})$ is the Fourier transform of the real-space two-body potential $U(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2)$. The operators $\hat{a}_{\mathbf{k}}^\dagger$ and $\hat{a}_{\mathbf{k}}$ represent the creation and destruction operators for the corresponding momentum eigenstates.

In the idealized limit where interactions are completely absent, the ground state of the system is trivial: all N bosons condense into the lowest-energy single-particle state featuring zero momentum ($\mathbf{k} = 0$). This macroscopic formation of a condensate is mathematically expressed by acting N times with the zero-momentum creation operator $\hat{a}_0^\dagger$ on the vacuum state $|0\rangle$ (incorporating the standard $1/\sqrt{N!}$ normalization factor for formal completeness):

$$|\psi_0^{(0)}\rangle = \frac{(\hat{a}_0^\dagger)^N}{\sqrt{N!}} |0\rangle. \quad (1.1.2)$$

Remark. *This clustering phenomenon stands in stark contrast to the behavior of a gas of fermions. Governed by the Pauli exclusion principle, fermions are strictly forbidden from occupying identical quantum states. Consequently, even at zero temperature, non-interacting fermions must sequentially fill all available momentum states up to a maximum threshold known as the Fermi energy, precluding any analogous zero-momentum condensation.*

We now introduce the principal assumption of our approach: the interparticle interaction is sufficiently weak. Given this, we anticipate that the true, interacting ground state remains a modest perturbation of the unperturbed condensate state $|\psi_0^{(0)}\rangle$. Therefore, even in the presence of interactions, the vast majority of the bosons will remain in the zero-momentum single-particle state. This physical condition is formalized by the inequality

$$\frac{N - N_0}{N} \ll 1, \quad (1.1.3)$$

where $N_0 = \langle \hat{a}_0^\dagger \hat{a}_0 \rangle$ defines the thermodynamic average number of bosons residing in the condensate. The quantity $N - N_0$ thus counts the number of depleted bosons that have been excited out of the condensate into states with $\mathbf{k} \neq 0$. Due to the weakness of the interaction, these non-zero momentum states are only microscopically occupied, meaning their occupation numbers scale as

<https://arxiv.org/pdf/1812.10732v5>.

$\langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle = \mathcal{O}(1)$. Crucially, the population of these excited states does not scale proportionally with the macroscopic total particle number N .

We leverage this assumption to implement a fundamental approximation on the interaction term of the Hamiltonian (1.1.1). Because operators acting on the condensate ($\hat{a}_0$ and $\hat{a}_0^\dagger$) scale macroscopically with $\sqrt{N_0} \approx \sqrt{N}$, we can safely neglect all interaction terms containing more than two operators equipped with non-zero momentum ($\hat{a}_{\mathbf{k}}$ or $\hat{a}_{\mathbf{k}}^\dagger$ for $\mathbf{k} \neq 0$). (Note that terms containing exactly three zero-momentum operators are inherently forbidden by momentum conservation requirements).

The most substantial and relevant term that survives this truncation is the one containing exclusively operators acting on the condensate ($\mathbf{q} = 0$ and $\mathbf{k} = \mathbf{k}' = 0$):

$$\frac{1}{2V} U(\mathbf{q} = 0) \hat{a}_0^\dagger \hat{a}_0^\dagger \hat{a}_0 \hat{a}_0 = \frac{1}{2V} U(\mathbf{q} = 0) (\hat{N}_0^2 - \hat{N}_0).$$

In the thermodynamic limit, $\hat{N}_0$ diverges linearly with N . Thus, we can completely disregard the trailing $\hat{N}_0$ term in the above expression, as it scales infinitely slower than the dominant quadratic term $\hat{N}_0^2 \sim N^2$.

To systematically express this leading-order contribution in terms of the conserved total particle number, we recognize that the number of bosons in the ground state can be written as the difference between the total number operator $\hat{N}$ and the sum of occupation numbers for all excited states:

$$\hat{N}_0 = \hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}},$$

where the primed summation specifically denotes a sum over all allowed momenta excluding zero. Letting $U_0 \equiv U(\mathbf{q} = 0)$, we substitute this relation back into our dominant interaction term. Expanding the square and purposefully discarding the resulting quartic term (which contains four out-of-condensate operators and is therefore negligible under our approximation), we obtain

$$\frac{1}{2V} U_0 (\hat{N}_0)^2 = \frac{1}{2V} U_0 \left(\hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right)^2 \approx \frac{1}{2V} U_0 \left(\hat{N}^2 - 2\hat{N} \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right).$$

Through this procedure, we successfully reduce the pure zero-momentum interaction into a quadratic form that couples the macroscopic total number N with the single-particle excitations.

Quadratic Interaction Terms and the Short-Range Assumption

Besides the dominant term where all operators are in the condensate state, we retain six terms from the interaction Hamiltonian that contain exactly two operators with momentum different from zero:

1. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k}' - \mathbf{q} = 0$, we have that $\mathbf{k} = -\mathbf{q}$ and $\mathbf{k}' = \mathbf{q}$, and so we remain with

$$\frac{1}{2V} a_0^\dagger a_0^\dagger \sum_{\mathbf{q}}' U(\mathbf{q}) a_{\mathbf{q}} a_{-\mathbf{q}},$$

where the sum $\sum_{\mathbf{q}}'$ is over all possible momenta $\mathbf{q} \neq 0$ because the zero case has been already taken into account in the previous purely macroscopic term.

At this stage, we introduce a crucial physical approximation to replace the interaction potential $U(\mathbf{q})$ with a simpler constant, $U(0)$. If we assume that the potential is sufficiently smooth, monotonic, and effectively short-ranged, its Fourier transform will be approximately constant for small momenta. Consequently, the main contribution to the interaction term comes from the zero-momentum component of the potential. In the context of two-body interactions, van der Waals forces serve as an excellent example of a short-ranged potential. In the standard Lennard-Jones potential, the attractive part is proportional to $1/r^6$ and the repulsive part to $1/r^{12}$. We could also consider dipole interactions, which scale as $1/r^3$; this represents a tricky marginal case for the short-range assumption. However, since we are primarily dealing with dilute atomic gases, van der Waals interactions are effectively short-ranged because the typical interparticle distance is much larger than the atomic radius. With this assumption, we can set $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$, making calculations significantly simpler. The first term thus becomes:

$$\frac{1}{2V} U_0 a_0^\dagger a_0^\dagger \sum_{\mathbf{q}}' a_{\mathbf{q}} a_{-\mathbf{q}}.$$

Remark. *As an analogous example of effective short-range interactions from high-energy physics, the strong nuclear force is also effectively short-ranged because its mediators, pions, are massive.*

2. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k}' = 0$, using again the short-range assumption, the term takes the following form:

$$\frac{1}{2V} U_0 \sum_{\mathbf{q}}' a_0^\dagger a_{-\mathbf{q}}^\dagger a_0 a_{-\mathbf{q}} = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{q}}' a_{-\mathbf{q}}^\dagger a_{-\mathbf{q}}.$$

3. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k} = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 \sum_{\mathbf{k}'}' a_0^\dagger a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{k}'}' a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}.$$

4. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k}' = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 \sum_{\mathbf{k}}' a_{\mathbf{k}}^\dagger a_0^\dagger a_0 a_{\mathbf{k}} = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{k}}' a_{\mathbf{k}}^\dagger a_{\mathbf{k}}.$$

5. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_0^\dagger a_{\mathbf{q}} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_{\mathbf{q}}.$$

6. If $\mathbf{k}' = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger a_0 a_0 = \frac{1}{2V} U_0 a_0 a_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger.$$

Condensate Mean-Field Approximation and the Bogoliubov Hamiltonian

A further crucial step is the replacement of the operators a_0 and $a_0^\dagger$ with numbers. Because the non-interacting ground state exhibits macroscopic occupation, its expectation value is $\langle a_0^\dagger a_0 \rangle =$

TODO: change: before the first bullet point, we should introduce the short-range assumption and explain why it is reasonable, and the result is that it has to decay faster than 1 over r cube.

$N_0 \approx N$. Therefore, we can confidently approximate $a_0 \sim \sqrt{N}$ and $a_0^\dagger \sim \sqrt{N}$ across all the previously derived terms of the approximate Hamiltonian that contain two condensate operators multiplied by two out-of-condensate operators.²

This mean-field approximation is formally justified by inspecting the commutator $[a_0, a_0^\dagger] = 1$ and comparing its magnitude to the result of acting with these operators on the non-interacting N -particle ground state $|\psi_0^{(0)}(N)\rangle$. Specifically, one finds:

$$a_0 |\psi_0^{(0)}(N)\rangle = \sqrt{N} |\psi_0^{(0)}(N-1)\rangle \sim \sqrt{N} |\psi_0^{(0)}(N)\rangle,$$

and

$$a_0^\dagger |\psi_0^{(0)}(N)\rangle = \sqrt{N+1} |\psi_0^{(0)}(N+1)\rangle \sim \sqrt{N} |\psi_0^{(0)}(N)\rangle,$$

where we have safely neglected 1 relative to the macroscopically large number N in these expressions. Because of this scaling, the value of the commutator is entirely negligible, and the operators can be effectively treated as commuting c-numbers.

Furthermore, during this systematic expansion, it is important to recognize why linear terms in $a_{\mathbf{k}}$ or $a_{\mathbf{k}}^\dagger$ do not appear. The terms which cancel out are precisely the ones that contain a single $\hat{a}_0$ or $\hat{a}_0^\dagger$ in the interaction term; if they did not cancel, they would have contributed to an unphysical renormalization of the free term of the Hamiltonian.

By substituting these numerical expectations back into our aggregated interaction terms, the full Hamiltonian becomes:

TODO: expand computations.

$$H = \sum_{\mathbf{k}} (\epsilon_k + nU_0) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{nU_0}{2} \sum_{\mathbf{k}} (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0, \quad (1.1.4)$$

where $n = N/V$ is the particle density.

Note that the second sum in the above equation contains the “anomalous” terms $a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger$ and $a_{\mathbf{k}} a_{-\mathbf{k}}$. Because the Hamiltonian now pairs creation operators together (and annihilation operators together) rather than consisting purely of standard number operators, it is not diagonal. To resolve this and find the true elementary excitations, one must employ the Bogoliubov transformation to diagonalize the Hamiltonian.

As a final remark, we note that if one were to carry out this derivation without invoking the constant potential approximation $U(\mathbf{q}) \simeq U_0$, the resulting Hamiltonian would naturally retain the momentum dependence of the interaction:

$$H = \sum_{\mathbf{k}} (\epsilon_k + nU(k)) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{n}{2} \sum_{\mathbf{k}} U(k) (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0.$$

TODO: add summary of assumptions: massive, weakly interacting, short-range, etc.

1.1.1 | Bogoliubov Transformations

We wish to diagonalize the Hamiltonian of the weakly interacting Bose gas by introducing new bosonic annihilation and creation operators, $\alpha_{\mathbf{k}}$ and $\alpha_{\mathbf{k}}^\dagger$, known as Bogoliubov operators. We need

²This can be pretty disturbing, since the expectation value of a single ladder operator is usually zero; but, as we will see later, we are not using eigenstates of number operator, instead superpositions of those (similar to coherent states). Indeed, even if the hamiltonian conserves the number of particle, the ground state description does not really rely on a fixed occupation of bosons. We will stress this point later on.

to construct a Hamiltonian which, in the end, depends only on a diagonal number operator of the form $\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$, yielding a target structure like:

$$H = \sum_{\mathbf{k}} E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \text{const},$$

for some energy dispersion E_k . To eliminate the anomalous terms that mix creation and annihilation operators in our previous Hamiltonian, we can perform a Bogoliubov transformation, which is a linear transformation mapping the original particle operators to the new quasiparticle operators. We start with the following standard ansatz for the Bogoliubov transformation:

$$\begin{cases} \alpha_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, \\ \alpha_{\mathbf{k}}^\dagger = v_k^* a_{\mathbf{k}}^\dagger + u_k^* a_{-\mathbf{k}}, \end{cases} \quad (1.1.5)$$

for $\mathbf{k} \neq 0$, where u_k and v_k are real coefficients. We now assume that these coefficients are real for simplicity, but they could in principle be complex. These coefficients are defined to depend strictly on the momentum magnitude $k = |\mathbf{k}|$, ensuring that rotational invariance is preserved in momentum space.

Because these newly defined operators must describe valid bosonic excitations, they are strictly required to satisfy the standard canonical bosonic commutation relations. We can verify this by evaluating the commutator between two annihilation operators:

$$\begin{aligned} [\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}'}] &= [u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, u_{k'} a_{\mathbf{k}'} + v_{k'} a_{-\mathbf{k}'}^\dagger] \\ &= u_k v_{k'} [a_{\mathbf{k}}, a_{-\mathbf{k}'}^\dagger] + v_k u_{k'} [a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}'}] \\ &= u_k v_{k'} \delta_{\mathbf{k}, -\mathbf{k}'} - v_k u_{k'} \delta_{-\mathbf{k}, \mathbf{k}'} = 0, \end{aligned}$$

which vanishes as expected. Next, we evaluate the critical commutator between a creation and an annihilation operator:

$$\begin{aligned} [\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}'}^\dagger] &= [u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, u_{k'} a_{\mathbf{k}'}^\dagger + v_{k'} a_{-\mathbf{k}'}] \\ &= u_k u_{k'} [a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] + v_k v_{k'} [a_{-\mathbf{k}}^\dagger, a_{-\mathbf{k}'}] \\ &= (u_k^2 - v_k^2) \delta_{\mathbf{k}, \mathbf{k}'}. \end{aligned}$$

To enforce the correct commutation relations, we must impose a hyperbolic normalization condition on the coefficients:

$$|u_k|^2 - |v_k|^2 = 1. \quad (1.1.6)$$

Provided this condition holds, the transformation linking the old and new operators can be easily inverted. We can express the linear mapping and its Hermitian conjugate in matrix notation:

$$\begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ v_k & u_k \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix}.$$

Because the determinant of the transformation matrix is exactly $u_k^2 - v_k^2 = 1$, the matrix is trivially invertible:

$$\begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & -v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{-\mathbf{k}}^\dagger \end{pmatrix},$$

which explicitly yields the inverted transformation:

$$\begin{cases} a_{\mathbf{k}} = u_k \alpha_{\mathbf{k}} - v_k \alpha_{-\mathbf{k}}^\dagger, \\ a_{\mathbf{k}}^\dagger = u_k \alpha_{\mathbf{k}}^\dagger - v_k \alpha_{-\mathbf{k}}. \end{cases}$$

We now proceed with the formal diagonalization of the Hamiltonian by expressing it entirely in terms of the new Bogoliubov operators:

$$\begin{aligned} H = & \sum_{\mathbf{k}}' [(\epsilon_k + nU_0)v_k^2 - nU_0u_kv_k] + \frac{1}{2}Vn^2U_0 \\ & + \sum_{\mathbf{k}}' [(\epsilon_k + nU_0)(u_k^2 + v_k^2) - 2nU_0u_kv_k] \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \\ & + \sum_{\mathbf{k}}' \left[\frac{nU_0}{2}(u_k^2 + v_k^2) - u_kv_k(\epsilon_k + nU_0) \right] (\alpha_{\mathbf{k}}^\dagger \alpha_{-\mathbf{k}}^\dagger + \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}}). \end{aligned}$$

It may initially appear that we have not made significant progress, as the final line still contains non-diagonal anomalous terms. However, we now have the mathematical freedom to choose arbitrary coefficients that eliminate these specific terms. By selecting an appropriate v_k (which simultaneously determines u_k via the constraint $u_k^2 = 1 + v_k^2$), we can force the bracketed prefactor of the off-diagonal terms to vanish identically:

$$\frac{nU_0}{2}(u_k^2 + v_k^2) - u_kv_k(\epsilon_k + nU_0) = 0.$$

Through simple algebraic manipulation and applying our normalization condition, this requirement simplifies to a second-order polynomial equation for v_k^2 :

$$4E_k^2v_k^4 + 4E_k^2v_k^2 - (nU_0)^2 = 0,$$

where we have suggestively defined the factor E_k^2 as:

$$E_k^2 = (\epsilon_k + nU_0)^2 - (nU_0)^2. \quad (1.1.7)$$

Solving this equation for v_k^2 directly yields the functional form of the transformation coefficients:

$$v_k^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right),$$

and consequently,

$$u_k^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} + 1 \right),$$

which implies for the cross-term:

$$u_kv_k = \frac{nU_0}{2E_k}.$$

By inserting these rigorously derived coefficients back into our expanded operator expression, the off-diagonal terms strictly cancel, yielding the final form of the diagonalized Hamiltonian:

$$H = \sum_{\mathbf{k}}' E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \mathcal{E}_0, \quad (1.1.8)$$

where $\mathcal{E}_0$ is a macroscopic constant energy shift defined by:

$$\mathcal{E}_0 = \frac{1}{2}Vn^2U_0 - \frac{1}{2} \sum_{\mathbf{k}}' (\epsilon_k + nU_0 - E_k).$$

The quantity $E_k = [(\epsilon_k + nU_0)^2 - (nU_0)^2]^{1/2} > 0$ represents the fundamental quasiparticle excitation spectrum of the system.

Non Interacting Limit

In the limit of vanishing interactions, $U_0 \rightarrow 0$, the Bogoliubov spectrum reduces to the free-particle dispersion: ...

TODO: add E and Eps in non interacting limit.

Ground State

With the Hamiltonian successfully diagonalized, we first seek its ground state, denoted $|\psi_0\rangle$. This state is mathematically defined as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$ for all non-zero momenta:

$$\alpha_{\mathbf{k}}|\psi_0\rangle = 0, \quad \forall \mathbf{k} \neq 0,$$

since any mode added to the vacuum would necessarily increase the energy of the system.

The corresponding energy eigenvalue of this ground state is simply the zero-point constant $\mathcal{E}_0$:

$$H|\psi_0\rangle = \mathcal{E}_0|\psi_0\rangle.$$

By acting with the creation operator $\alpha_{\mathbf{k}}^\dagger$ on this vacuum state, we can construct the entire spectrum of excited states:

$$\frac{(\alpha_{\mathbf{k}_1}^\dagger)^{n_{\mathbf{k}_1}}}{\sqrt{n_{\mathbf{k}_1}!}} \frac{(\alpha_{\mathbf{k}_2}^\dagger)^{n_{\mathbf{k}_2}}}{\sqrt{n_{\mathbf{k}_2}!}} \cdots \frac{(\alpha_{\mathbf{k}_N}^\dagger)^{n_{\mathbf{k}_N}}}{\sqrt{n_{\mathbf{k}_N}!}} |\psi_0\rangle,$$

whose corresponding energy eigenvalues are given by:

$$\mathcal{E}_0 + \sum_{i=1}^N n_{\mathbf{k}_i} E_{\mathbf{k}_i},$$

where $n_{\mathbf{k}_i} = \alpha_{\mathbf{k}_i}^\dagger \alpha_{\mathbf{k}_i}$ is the occupation number of the i -th state in the *Bogoliubov quasiparticle basis*.

It is critical to note that the vacuum of these newly defined operators, $|\psi_0\rangle$, is distinct from the vacuum of the original bare particle operators, $a_{\mathbf{k}}$. This distinction becomes physically apparent when we evaluate the expectation value of the bare particle number operator in the interacting ground state:

$$\begin{aligned} \langle \psi_0 | a_{\mathbf{k}}^\dagger a_{\mathbf{k}} | \psi_0 \rangle &= \langle \psi_0 | (u_{\mathbf{k}} \alpha_{\mathbf{k}}^\dagger - v_{\mathbf{k}} \alpha_{-\mathbf{k}}) (u_{\mathbf{k}} \alpha_{\mathbf{k}} - v_{\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger) | \psi_0 \rangle \\ &= \langle \psi_0 | \left[u_{\mathbf{k}}^2 \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + v_{\mathbf{k}}^2 \alpha_{-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger - u_{\mathbf{k}} v_{\mathbf{k}} (\alpha_{\mathbf{k}}^\dagger \alpha_{-\mathbf{k}}^\dagger + \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}}) \right] | \psi_0 \rangle \\ &= v_{\mathbf{k}}^2 \neq 0, \end{aligned}$$

where the final evaluation leverages the commutation relation $[\alpha_{-\mathbf{k}}, \alpha_{-\mathbf{k}}^\dagger] = 1$ and the fact that annihilation operators destroy the vacuum. This previous calculation elegantly demonstrates a profound physical consequence of the theory: *even in the absolute zero-temperature ground state $|\psi_0\rangle$, the mean occupation number of states with non-zero momentum is strictly positive*. In terms of the original operators, we would state that $n_{\mathbf{k}} \neq 0$ (where $n_{\mathbf{k}} = a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$) even when $\mathbf{k} \neq 0$. Because of the interactions, a finite fraction of bosons is fundamentally pushed out of the zero-momentum state, leading to a phenomenon known as quantum depletion, where the condensate is only partially occupied even at zero temperature.

1.2 | Condensate Depletion and Excitations

We now want to compute the non condensate fraction of the weakly interacting Bose gas, which is given by the expectation value of the number operator for non-zero momentum states:

$$N - N_0 = \sum_{\mathbf{k}} \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle,$$

recalling that the expectation value of $n_{\mathbf{k}}$ (number operator of the Bogoliubov quasiparticles) is $v_{\mathbf{k}}^2$, which in the non interacting limit goes to zero:

$$v_{\mathbf{k}}^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right) \xrightarrow{U_0 \rightarrow 0} 0.$$

[...]

sum to integrals

[...]

We will just check the convergence of the integral in the ultraviolet limit ($k \rightarrow \infty$), and in the infrared limit ($k \rightarrow 0$).

[...]

dimensionless variables and occupation

$$n' = \dots$$

[...]

We can thus interact with the condensate by operating on the density, the only free parameter in $c = \sqrt{\frac{nU_0}{m}}$.

TODO: add graph of ground state occupation number

Example (Scattering Length). The scattering length a is a measure of the strength of the interaction between two particles. It is defined as the distance at which the wave function of two interacting particles goes to zero. In the context of Bose-Einstein condensates, the scattering length plays a crucial role in determining the properties of the condensate, including its stability and the nature of its excitations.

For weakly interacting Bose gases, the scattering length can be related to the interaction potential U_0 through the relation:

$$U_0 = \frac{4\pi\hbar^2 a}{m},$$

where m is the mass of the particles. This relation allows us to express various physical quantities in terms of the scattering length, providing a more intuitive understanding of how interactions affect the condensate.

[...]

condensate depletion computation with na

comment for which the result does not have any $\hbar$ term or terms we can expand to any classical or semiclassical limit. This is a consequence of the fact that the depletion and BEC in general are purely quantum effects, arising from the interactions between particles in the condensate.

[...]

1.3 | Ground State Energy

[...]

Pressure and Speed of Sound

[...]

1.3.1 | Ground State Wave Function

[...]

1.4 | Spontaneous Symmetry Breaking

[...]

1.5 | Specific Heat and Condensate Fraction

Let us proceed with the calculation of the specific heat of a dilute Bose gas.

[...]

$$\begin{aligned}\frac{E(T)}{V} &= \dots \\ &= \frac{\mathcal{E}_0}{V} + \frac{\pi^2}{30} \frac{(k_B T)^4}{(c\hbar)^3},\end{aligned}$$

[...]

$$C_V = \dots$$

Exercises

Sheet 1

[...]

Appendices

A | Notation and Conventions

B | Computations and further developments